

Space Plasmas in the Solar System, including Planetary Magnetospheres (D)
Plasma Transport across Magnetospheric Boundaries (D3.7)
Consider for oral presentation.

A KP FORECAST TO FIVE DAYS IN ADVANCE FROM SOLAR SURFACE EX- TRAPOLATED GSM COMPONENT FIELDS AND HELIOSPHERIC-DERIVED VELOCITY AND DENSITY

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Our UCSD group now provides a Kp forecast up to five days ahead of the current time with a 70% chance of an occurrence rate to predict geomagnetic storms from a Kp enhancement greater than 5. We provide this from first principles (Newell et al., doi: 10.1029/2006JA012015) using a machine learning tool and a prediction of GSM magnetic field components, velocity, and density. This forecast is currently made available on the UCSD website <https://ips.ucsd.edu> and to the NASA Goddard Community Coordinated Modeling Center. Our automatic system operates using near-Earth spacecraft measurements and Interplanetary Scintillation (IPS) data from existing world radio sites provides the density and velocity forecasts. Magnetic fields using Global Oscillation Network (GONG) data sets provide GSM fields at Earth extrapolated outward from the solar surface. We have known since 2018 (Jackson et al., doi: 10.1029/2018SW002098) that we were able to forecast GSM Bz fields many days ahead. However, since the summer of 2024 our machine learning tool has been used to provide the high Kp correlation with geomagnetic storms in advance of our observations. We show past examples of real time forecasts in this presentation.